Linear Algebra | Quiz for §4.6 | October 30, 2026

Name: _____ R. Hammack Score: _____

Directions: Show and explain all work. Put phones away. No calculators, etc.

This quiz concerns the matrix $A = \begin{bmatrix} 3 & 3 & 15 & 11 \\ 1 & -3 & 1 & 1 \\ 2 & 3 & 11 & 11 \end{bmatrix}$

1. Find a basis for the row space of A.

   Begin by putting A into reduced row echelon form.

   $$\begin{bmatrix} 3 & 3 & 15 & 11 \\ 1 & -3 & 1 & 1 \\ 2 & 3 & 11 & 11 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -3 & 1 & 1 \\ 3 & 3 & 15 & 11 \\ 2 & 3 & 11 & 11 \end{bmatrix} \xrightarrow{\substack{R_2 - 3R_1 \to R_2 \\ R_3 - 2R_1 \to R_3}} \begin{bmatrix} 1 & -3 & 1 & -3 \\ 0 & 12 & 12 & 8 \\ 0 & 9 & 9 & 9 \end{bmatrix}$$

   $$\xrightarrow{\substack{\frac{1}{4}R_2 \to R_2 \\ \frac{1}{9}R_3 \to R_3}} \begin{bmatrix} 1 & -3 & 1 & -3 \\ 0 & 3 & 3 & 2 \\ 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{R_1 + 3R_3 \to R_1 \\ R_2 - 3R_3 \to R_2}} \begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

   $$\xrightarrow{R_3 + R_2 \to R_3} \begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{-R_2 \to R_2 \\ R_2 \leftrightarrow R_3}} \begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = B \text{ (reduced row echelon form)}$$

   Now we can read off a basis for the row space from B: $\boxed{\left\{ [\,1\,0\,4\,0\,],\ [\,0\,1\,1\,0\,],\ [\,0\,0\,0\,1\,] \right\}}$

2. Find a basis for the column space of A.

   The second column of B is a linear combination of the first two columns.

   The remaining columns (first, second and fourth) of B are linearly independent,

   so the corresponding columns of A form a basis for the column space of A.

   Answer: $\boxed{\text{A basis for the column space of } A \text{ is } \left\{ \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}, \begin{bmatrix} 11 \\ 1 \\ 11 \end{bmatrix} \right\}.}$

3. Find the dimension of the null space of A, that is, find $\text{nulity}(A)$.
   (You may do this either by finding a basis for $\text{Null}(A)$, or by using the Rank-Nullity Theorem.)

   For an $m \times n$ matrix A, the Rank-Nullity Theorem states that $n = \text{rank}(A) + \text{nullity}(A)$.

   For the matrix A in question, this becomes $4 = 3 + \text{nullity}(A)$, so $\boxed{\text{nullity}(A) = 1}$.

   Alternatively, you could argue as follows. The null space of A is the set of solutions to $A\mathbf{x} = \mathbf{0}$, which is the same as the set of solutions to $B\mathbf{x} = \mathbf{0}$, or $\begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

   This corresponds to the system $\begin{cases} w + 4y = 0 \\ x + y = 0 \\ z = 0 \end{cases}$, or $\begin{cases} w = -4y \\ x = -y \\ y = \text{(free)} \\ z = 0 \end{cases}$

   So the solution is the set of all vectors $\begin{bmatrix} -4y \\ -y \\ y \\ 0 \end{bmatrix} = y \begin{bmatrix} -4 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \text{Span}\left(\begin{bmatrix} -4 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right)$.

   Consequently the column space is 1-dimensional, so $\boxed{\text{nullity}(A) = 1}$.